



DATABASE AND BACKEND MANAGEMENT — Section 10

Event and Room Occupancy Management System

A centralized web platform for smarter room and event management on campus

TEAM - 1

2520090123 • Jalla Geya Reddy

2520090001 • Kamtam Priyanka

Problem Statement



01

Finding Available Rooms Is Difficult

Students frequently struggle to locate free rooms during college hours, since there is no shared, up-to-date view of room availability.



02

Spot Registrations Disrupt Planning

Hackathons, workshops, seminars, and competitions often receive extra participants through spot registrations, pushing attendance beyond the expected count.



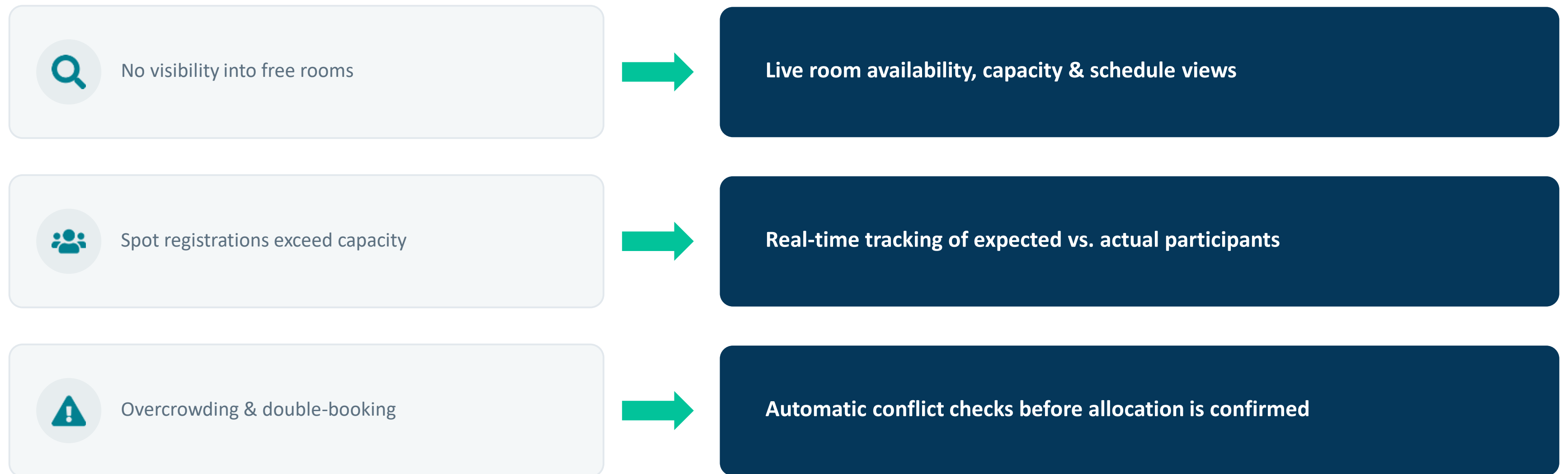
03

Overcrowding & Conflicts Follow

These gaps lead to overcrowding, room allocation clashes, scheduling conflicts, and inefficient use of limited college spaces.

Proposed Solution

We propose to design and develop a centralized, web-based platform that will bring room availability and event management onto a single system — replacing scattered, manual coordination.



Key Features

The proposed system will provide role-based functionality for two types of users.



Student / User Features

- View available and occupied rooms in real time
- Check room capacity and schedules
- Browse and register for college events
- Track expected vs. registered participant counts

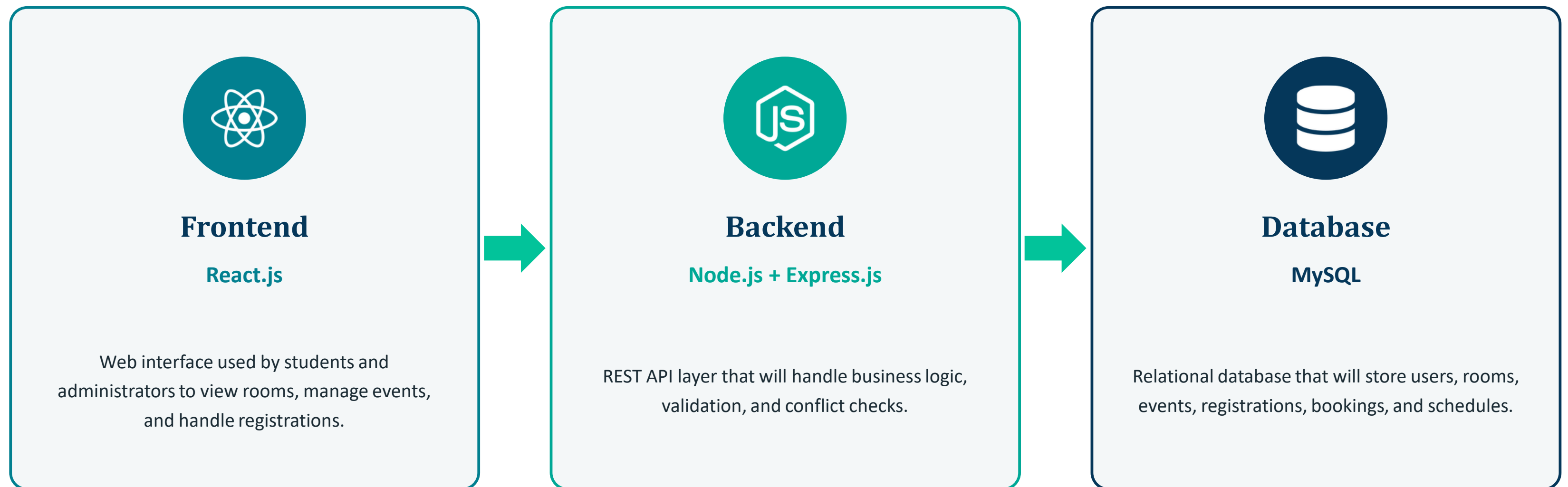


Administrator Features

- Manage room bookings and allocations
- Create and manage college events
- Update and monitor actual occupancy
- Detect scheduling conflicts and block allocation when capacity or timing clashes occur

System Architecture

The proposed system will follow a standard three-layer web architecture.



Supporting Tools: MySQL Workbench (design) • Postman (API testing) • VS Code (development) • Git & GitHub (version control)

Expected Outcome & Future Scope



Better utilization of college rooms and spaces



Reduced scheduling conflicts and double-bookings



Improved handling of event registrations and occupancy



A single, centralized system for room and event management



Future Scope

Beyond the core system, future iterations may explore automated notifications for schedule changes, analytics dashboards for space utilization trends, and reporting tools to support long-term campus planning.